

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“App Development and ML Model Training Project for Foot Pressure Analyzer Device”

Submitted in partial fulfillment for the award of degree(20INT82)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

S B Gagan - 1DS22AI404

Conducted at



**SPV4 Ameya Innovation Labs Private
Limited**

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

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CERTIFICATE

This is to certify that the Internship titled “**App Development and ML Model Training Project for Foot Analyzer Device**” carried out by **Mr. S B GAGAN [1DS22AI404]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering, in Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)

Signature of Reporting Head
Dr. Sandip Kumar Roy
Director
SPV4 Ameya

Signature of Guide
Dr. Aruna M G
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

Signature with Date

1) _____

2) _____

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **S B Gagan**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **SPV4 Ameya Innovation Labs Private Limited**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date:

USN: 1DS22AI404

Place: Bangalore

NAME: S B Gagan

OFFER LETTER



SP4 AMEYA INNOVATION LABS PRIVATE LIMITED

#startupindia



OFFER LETTER OF INTERNSHIP

Date: 07/09/2024

Ref# SP4Ameya/Ltr2024/21

Sub: Internship Offer - App Development and ML Model Training Project

Dear S B Gagan,

We are pleased to offer you the 14-15 weeks internship position for the "App Development and ML Model Training Project" associated with our Foot Pressure Analyzer Device. Your skills and interests align well with the project requirements, and we are excited to have you on board.

We look forward to your acceptance and a fruitful collaboration.

Yours Faithfully,

A handwritten signature in blue ink, appearing to read "Dr. Sandip Kumar Roy".

Dr. Sandip Kumar Roy
(Director)

DIN: 10532220

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. I take this opportunity to express my gratitude to all who have helped us to complete the Internship.

I express my sincere thanks to our Principal **Dr. B G Prasad**, for providing me adequate facilities to undertake this Internship.

I would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing me an opportunity to carry out Internship and for her valuable guidance and support.

I would like to thank **Dr. Sandip Kumar Roy**, Director, SPV4 Ameya Innovation Labs Private Limited for guiding me during the period of internship.

I express my deep and profound gratitude to my Guide, **Dr. Aruna M G**, Associate Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

I would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

I would like to thank the non-teaching members of the Department of AI & ML, for helping me during the Internship.

Last but not the least, I would like to thank my parents and friends without whose constant help, the completion of Internship would have not been possible.

S B GAGAN [1DS22AI404]

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CHAPTER 1

INTRODUCTION

INTRODUCTION TO FOOT PRESSURE ANALYZER DEVICE AND APP

Foot pressure analyzer device is an essential diagnostic tool used by healthcare professionals to evaluate foot function, gait patterns, and balance. By measuring the distribution of pressure across different areas of the foot during standing or walking, this method helps in detecting abnormalities, monitoring rehabilitation progress, and preventing injuries. It is particularly valuable for individuals with conditions like diabetes, arthritis, or those undergoing physiotherapy and post-surgical recovery.

To bridge the gap between clinical assessment and everyday monitoring, Safe Step, a smart mobile application that has been developed as a digital solution for real-time foot pressure analysis. The app is designed to work in conjunction with wearable foot pressure sensors and provides users with detailed insights into their step count, balance, and overall walking pattern.

Safe Step is particularly tailored for patients and doctors, offering personalized analytics, progress tracking, and communication features. By integrating Machine Learning (ML) algorithms, the app can detect deviations from normal walking patterns and alert users or caregivers to potential risks.

This internship project at SP4 Ameya Innovation Labs Private Limited focused on the development and enhancement of the Safe Step app, involving key tasks like Flutter-based mobile development, Firebase integration, and training ML models to interpret sensor data and generate meaningful feedback.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

SPV4 AMEYA INNOVATION LABS PRIVATE LIMITED



ABOUT THE COMAPNY

SP4 Ameya Innovation Labs Private Limited is a healthcare technology company focused on transforming the MedTech landscape through the development of smart, accessible, and efficient healthcare solutions. Established with the objective of bridging the gap between digital health transformation and patient-centric care, the organization specializes in creating scalable and affordable physiotherapy devices, wearable rehabilitation tools, and AI-integrated healthcare applications.

The company integrates advanced technologies such as Artificial Intelligence (AI), the Internet of Things (IoT), and cloud computing to offer solutions that address both current and future challenges in the healthcare ecosystem. With a strong emphasis on inclusivity, SP4 Ameya Innovation Labs designs its products and services to improve accessibility for individuals across all levels of physical ability and to enhance the operational efficiency of healthcare providers.

Innovations developed at SP4 Ameya are targeted towards improving therapeutic outcomes, increasing patient safety, and delivering real-time, data-driven insights for healthcare decision-making. Through collaborative research and development, the company continues to redefine standards in smart medical technology and rehabilitation science.

VISION:

To bridge the gap between healthcare needs and digital transformation by enabling accessible, intelligent, and personalized healthcare innovations that enhance quality of life and deliver patient-centric care.

MISSION:

To pioneer accessible healthcare innovations by leveraging AI-powered analytics, modular product designs, and IoT-enabled devices. The mission encompasses enhancing therapeutic efficiency, improving safety standards, and delivering scalable, affordable, and portable

solutions compatible with telehealth platforms. The company focuses on providing technology-driven tools that support predictive healthcare delivery and empower patients and providers alike.

CORE AREAS OF INNOVATION:

- **Wearable Rehabilitation Tools:**

Design and development of ergonomic, data-driven wearable devices that support physiotherapy and rehabilitation. These tools enable real-time monitoring of patient movements and therapy effectiveness, facilitating precise treatment planning.

- **AI-Integrated Applications:**

Development of mobile and web-based applications powered by artificial intelligence to support automated diagnosis, progress tracking, and personalized treatment recommendations. These applications are designed to improve engagement and outcomes for both patients and healthcare professionals.

- **IoT and Cloud-Based Infrastructure:**

Implementation of IoT-enabled devices connected through cloud platforms to ensure seamless data collection, synchronization, and access across multiple systems. This infrastructure supports real-time analytics and enhances the efficiency of healthcare delivery.

- **Modular and Portable Medical Devices:**

Creation of compact, customizable, and transportable devices suitable for use in clinical settings as well as at-home care environments. These devices are built to be cost-effective and scalable, ensuring accessibility across a wide user base.

- **Telehealth-Compatible Solutions**

Integration of telemedicine features into products and applications to facilitate remote healthcare delivery. This includes virtual consultations, remote monitoring, and digital health reporting, ensuring continuous care even in remote or underserved locations.

CHAPTER 3

SCOPE OF WORK

The scope of work during the internship at SP4 Ameya Innovation Labs Private Limited centered around active participation in the design, development, and testing processes of healthcare-focused mobile applications. The primary project handled was the Safe Step App—a smart foot pressure monitoring system that integrates machine learning with cloud-based infrastructure. Additionally, the internship involved testing an earlier product developed by the organization, providing comprehensive exposure to the software development life cycle (SDLC), quality assurance practices, and user-centric design in a MedTech setting.

1. Application Testing (Initial Phase)

The internship commenced with involvement in the quality assurance and testing of an existing mobile application built by the company. This phase laid the groundwork for understanding product development and contributed to proficiency in:

- Evaluating application flow and feature execution
- Detecting bugs and reporting issues methodically
- Reviewing UI/UX for functionality and accessibility
- Executing test cases and compiling structured feedback

Testing was primarily carried out on Android platforms, covering both functional and non-functional checks. Performance glitches and usability issues were reported using standardized templates, and enhancement suggestions were relayed to the development team. This phase offered valuable exposure to MedTech testing workflows and helped shape quality benchmarks for future projects.

2. Wireframe Contribution for Safe Step App

One of the significant contributions was toward the UI/UX wireframing of the Safe Step App, which is a mobile application designed to analyze foot pressure data collected from wearable sensors. The wireframe design was created using Figma, and efforts were directed toward:

- Designing intuitive and accessible user interfaces tailored for doctors and patients.
- Structuring app navigation flow to support ease of use and clarity in user interaction.
- Ensuring that the app reflected the company's vision of inclusivity, efficiency, and patient-centered care

- Providing design feedback and collaborating with developers for UI refinement

The wireframe played a critical role in setting the foundation for the final app design. Attention was given to accessibility, minimalism, and a clean layout to ensure a seamless user experience, especially for patients with limited digital literacy.

3. Frontend Development Using Flutter

Post wireframe design, the internship involved hands-on contribution to frontend development using the Flutter framework, which is known for its cross-platform capabilities and fast development cycle. Key tasks included:

- Assisting in the implementation of the splash screen and navigation logic based on Firebase authentication status
- Developing UI components and integrating interactive elements as per the wireframe
- Customizing buttons, input fields, and layout structures in alignment with the design vision
- Debugging interface issues and enhancing responsiveness across screen sizes

The frontend phase provided exposure to Flutter widgets, state management, routing, and platform-specific adaptations, thereby strengthening mobile development skills in a real-world project.

4. Backend Integration Using Firebase

In parallel with frontend work, efforts were directed toward integrating the app's backend using Firebase, which managed authentication, cloud data storage, and user-specific logic. Key tasks included:

- Configuring FirebaseAuth for secure user login and signup
- Structuring Firestore collections to organize patient and therapist records
- Enabling frontend-backend data exchange through Firestore operations
- Contributing to logic that determined user flow based on login status and profile completion

This integration ensured secure and scalable cloud-based operations for the Safe Step app, aligning with SP4 Ameya's commitment to IoT-ready, telehealth-compatible healthcare solutions.

The internship involved diverse tasks spanning app testing, UI/UX design, mobile frontend development, and cloud backend integration, enabling comprehensive involvement in the app development process. By contributing to both technical and design aspects of the Safe Step app, and engaging in the testing of a prior project, the internship offered hands-on exposure to real-world software development in the context of smart healthcare applications.

CHAPTER 4

LEARNING OBJECTIVES

1. Practical Experience in Cross-Platform App Development Using Flutter:

A key learning outcome was acquiring hands-on development experience with Flutter, a powerful cross-platform UI toolkit. Throughout the internship, involvement in building parts of the Safe Step mobile app allowed for practical learning in areas like widget customization, layout design, navigation management, and responsive UI development. This opportunity provided a strong foundation in modern app architecture, state handling, and design-to-code translation. The experience was instrumental in understanding how scalable apps are built for both Android and iOS platforms with a single codebase.

2. Implementing Backend Connectivity and Real-Time Features Using Firebase:

This internship emphasized mastering backend integrations using Firebase. The app required seamless data flow and secure user authentication, which involved working with Firebase Authentication and Firestore database. Gaining familiarity with features like data fetching, profile-based navigation, and real-time updates helped deepen the understanding of client-server interaction. This learning objective enhanced practical skills in asynchronous programming, cloud data operations, and maintaining secure communication between frontend and backend systems.

3. Collaborating in UI/UX Wireframe Design for Healthcare Application:

Participating in the wireframe design process provided valuable exposure to user interface planning and user experience thinking. Using Figma, the internship involved co-creating wireframes and mapping user journeys specific to patients and doctors. Emphasis was placed on ensuring intuitive layouts, efficient navigation, and healthcare accessibility—especially considering users with physical constraints. The exercise fostered a user-first mindset and taught the importance of translating medical functionality into a friendly digital experience. The process provided insight into how design decisions impact usability, and how to collaborate effectively with developers to bring those wireframes to life while maintaining the intended user experience.

4. Strengthening Skills in Testing, Debugging, and Quality Assurance:

Another critical area of learning was understanding software testing and debugging processes in real-time development cycles. Initially contributing as a QA tester for a different healthcare application allowed practice in identifying bugs, verifying UI consistency, and analyzing user journeys. This background proved useful when transitioning into debugging and refining the Safe Step app. The learning objective sharpened diagnostic skills and highlighted the importance of iterative development in maintaining code quality. This objective reinforced the importance of iterative development and quality assurance in professional app development.

5. Gaining Insight into Tech-Driven Solutions in Digital Health:

This internship served as a gateway to understanding the practical role of technology in modern healthcare delivery. Working on a foot pressure analysis application opened up learning about wearable sensor integration, physiotherapy-focused app design, and how data visualization aids clinical evaluation. Exposure to the intersection of medical needs and digital platforms helped appreciate the impact of AI and mobile technologies in rehabilitation and patient monitoring. This learning bridged the gap between theoretical knowledge and practical application, especially in understanding how digital health solutions can empower both patients and healthcare providers.

CHAPTER 5

CHALLENGES AND SOULTION

1. Implementing Smart Navigation Based on User Status:

Challenges:

The Safe Step app required a smart navigation mechanism that redirected users based on their authentication status and whether they had completed their profile and personal details. Integrating this logic within the splash screen using FirebaseAuth and Firestore presented early difficulties. Issues such as delayed screen transitions, incorrect redirections, or null values from the database led to inconsistent app behavior and disrupted the user experience.

Solution:

The navigation logic was refined by restructuring the asynchronous functions inside the initState() of the splash screen. Defensive programming techniques were used by implementing null checks and initializing fallback values. Boolean fields like profileCompleted and detailsCompleted were fetched from Firestore before determining the navigation route using a well-ordered conditional flow. Additionally, a short delay using Future.delayed allowed Firebase to fully initialize before proceeding. These optimizations made the navigation more reliable and fluid during startup.

2. Ensuring Smooth Real-Time Updates with Firestore:

Challenges:

Maintaining real-time synchronization of user data—like health parameters, steps, or profile completion—posed a challenge when using Firebase Firestore. The app sometimes failed to reflect immediate updates in the UI, or repeated entries occurred due to poor handling of data streams and concurrency issues.

Solution:

This was solved by leveraging Firebase's real-time features using StreamBuilder, which allowed the UI to respond instantly to backend changes. Firestore security rules were carefully reviewed to ensure consistent read/write access, and throttling mechanisms were applied to prevent excessive writes. For sensitive updates, Firestore transactions were implemented to maintain data consistency. These enhancements improved responsiveness and ensured accurate data representation across the app.

3. Designing Accessible and Functional Wireframes for Healthcare Use:

Challenges:

Creating a user-friendly interface for a healthcare-focused application posed several design challenges. With users potentially being elderly or undergoing physical therapy, the UI needed to be highly accessible, clear, and easy to navigate. The initial wireframes lacked the clarity and intuitive flow expected in a MedTech environment, which led to confusion during testing.

Solution:

The team adopted a user-centric approach by researching user personas and similar medical apps. Using Figma, wireframes were created with accessibility in mind—focusing on clear typography, high contrast visuals, and larger interactive elements for ease of use. Helpful features like dropdowns, iconography, and guided text were added to enhance user navigation. Continuous team feedback and iteration helped refine the wireframes into an interface that balanced aesthetics, usability, and clinical functionality.

HARDWARE AND SOFTWARE REQUIREMENTS

Software requirements:

- Flutter SDK
- Firebase
- Android Studio
- Visual Studio Code
- Figma
- Git & GitHub
- Firebase CLI (Command Line Interface)
- Google Chrome / Android Emulator

Hardware Requirements:

- Processor: Intel i5 or more
- RAM: 8 GB minimum
- Display: 23 LED Display
- 64bit operating systems
- Android Smartphone
- LAN Connection Port: Required

CHAPTER 6

ACHEIVEMENTS AND CONTRIBUTION

1. Initial QA & App Testing Phase:

At the onset of the internship, detailed manual testing was conducted on an existing mobile application developed by SP4 Ameya Innovation Labs for a different healthcare-related project. The focus was on identifying UI inconsistencies, navigation errors, and responsiveness issues across a variety of screen sizes and device configurations. All findings were meticulously documented in structured bug reports and shared with the development team for resolution. Beyond functional validation, usability feedback was offered, particularly targeting improvements in the onboarding flow to help new users get started with minimal friction. This phase not only elevated the overall app quality but also deepened the understanding of user-first QA practices and the iterative nature of refining digital health platforms.

2. Wireframe Ideation & UI Planning for Safe Step:

Active contributions were made to the design ideation phase of the Safe Step application, where input was provided during team discussions focused on building a simple, accessible, and visually coherent interface. Using Figma, the wireframe layout was collaboratively developed, addressing navigation flow, logical screen transitions, intuitive button placements, and onboarding experiences tailored for both patients and doctors. The designs placed a strong emphasis on inclusivity and accessibility, catering to users who may have physical limitations. Specific improvements were made to the home dashboard (featuring step tracking and gait insights) and the authentication flow. The doctor interface was also carefully structured to provide clear visibility into patient metrics, reports, and communication tools in an organized format.



Fig 6.1: Wireframe of Safe Step App

3. Front-End Development using Flutter:

Following the design phase, active involvement continued in transforming the wireframe designs into functional user interface screens using Flutter. This included working with a variety of Flutter widgets to bring the app's visual structure to life while maintaining consistency with the proposed layout. Efforts were made to ensure that the application followed responsive design principles, allowing it to render effectively across different screen sizes and orientations. Key contributions included supporting the creation of the splash screen, which featured the app's logo and incorporated FirebaseAuth initialization logic to determine user login status. Assistance was also provided in building the login and signup pages, complete with input validation and a streamlined user flow for seamless account creation. Navigation logic was implemented to enable conditional routing, allowing the app to dynamically redirect users based on their profile completion status. In addition, active debugging of UI issues and optimization of the codebase helped ensure clean, modular development practices, including the use of reusable widgets for better maintainability and scalability.

4. Firebase Integration:

Significant contributions were made to the backend integration of the Safe Step application by working extensively with Firebase services. Firebase Authentication was implemented to manage secure user login and signup functionality using email and password credentials, ensuring data protection and user account integrity. Cloud Firestore was utilized to store and retrieve vital user data such as personal profile information, steps taken, and other health-related metrics. To enhance user experience with live updates, real-time database listeners were integrated, enabling automatic synchronization between backend data changes and the app's user interface. Additionally, collaboration with senior developers facilitated the development of an efficient navigation wrapper that dynamically directed users based on their Firebase authentication and profile completion status. This seamless backend integration significantly contributed to making the app responsive, secure, and tailored to individual user journeys.



Fig 6.2: Firebase console depicting the authentication details of users

CHAPTER 7

REFLECTION AND ANALYSIS

The internship at SP4 Ameya Innovation Labs Private. Ltd. proved to be a transformative experience, offering comprehensive exposure to both mobile application development and machine learning integration within the healthcare sector. Being a part of the Safe Step project allowed for not just technical growth, but also a deeper appreciation of how thoughtfully engineered digital solutions can impact patient care and physiotherapy outcomes.

A key highlight of this internship was the hands-on participation in multiple stages of the development cycle, from wireframing and UI design to implementing core app functionalities using Flutter and integrating real-time features with Firebase. This end-to-end involvement fostered a stronger connection to the project and helped build practical skills across UI/UX design, secure authentication, data syncing, and cross-device compatibility.

The work culture at SP4 Ameya promoted open discussion and collaborative problem-solving, which nurtured a creative mindset. Engaging with teammates and mentors during design reviews and development sprints emphasized the role of user empathy, especially important in designing healthcare applications that are easy to use for patients with limited mobility or tech familiarity.

Addressing technical hurdles such as asynchronous data handling, conditional routing, and responsive layouts provided critical learning moments. These challenges highlighted the importance of clean, maintainable code and the value of rigorous testing to ensure a seamless user experience.

On a personal front, the internship greatly boosted proficiency in tools like Figma, Flutter, and Firebase, while also enhancing vital soft skills such as teamwork, communication, and analytical thinking, skills that are indispensable in today's fast-paced, interdisciplinary tech landscape.

CHAPTER 8

RECOMMENDATIONS

- 1. Expand Cross-Device Compatibility Testing:** To maintain a consistent user experience across all devices, it is essential to regularly test the Safe Step app on various screen sizes, resolutions, and OS versions. This practice will help catch design inconsistencies and performance bottlenecks early, ensuring reliability and accessibility across platforms.
- 2. Adopt Role-Based User Permissions:** Although the app supports basic patient and doctor profiles, implementing detailed role-based permissions will enhance data security and streamline user access. This ensures that healthcare professionals only view authorized patient data, while users retain control over their personal records.
- 3. Introduce In-App User Feedback Feature:** Embedding a feedback option within the app would enable users to instantly share their experiences, report bugs, or suggest features. Collecting this real-time input helps drive user-centric improvements and fosters greater transparency and engagement.
- 4. Refine the Onboarding Workflow:** While the current onboarding process is functional, it can be further enhanced with visual aids such as tooltips, walkthrough animations, or mini-guides. This approach simplifies the user journey, especially beneficial for elderly users or those new to mobile healthcare apps.
- 5. Build a Live Analytics Interface for Doctors:** A real-time data dashboard for doctors can display trends in patient steps, gait patterns, and activity levels. This feature can support early intervention strategies, enhance monitoring, and provide data-driven insights for better healthcare outcomes.

CHAPTER 9

CONCLUSION

The internship at SP4 Ameya Innovation Labs Pvt. Ltd. has been an immensely rewarding journey that provided practical exposure to the intersection of mobile app development and machine learning in the healthcare sector. Working on the Safe Step application enabled hands-on experience with industry tools like Flutter, Firebase, and Figma, while reinforcing the value of user-centered thinking in building effective digital health solutions.

Being involved across multiple stages of the project—from initial QA testing of a pre-existing healthcare app to participating in UI/UX ideation, wireframe creation, frontend development, and backend integration—contributed to a holistic learning experience. Each task helped develop core technical skills, sharpen attention to detail, and cultivate effective communication within a collaborative team environment.

The internship deepened the understanding of how healthcare technologies must balance functionality, data protection, and simplicity. Emphasis on responsive design, role-based access, and patient-doctor interfaces underscored the need for accessible platforms tailored to users with diverse needs, including those with mobility impairments.

The fast-paced, startup-like work environment encouraged proactive learning and flexible problem-solving. Regular feedback from mentors and collaboration with experienced developers bridged the gap between theoretical knowledge and real-world implementation, especially in areas such as real-time data handling, modular coding, and secure authentication systems.

Overall, this experience not only strengthened practical knowledge in app architecture and backend services but also sparked a genuine passion for health-tech innovation. It has solidified the desire to build technology that is not only efficient but also empathetic, driving positive change in how healthcare is delivered and experienced.

CHAPTER 10

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